



OPEN Mapping neighbourhood-level drivers of type 2 diabetes for precision public health using predictive and causal machine learning

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Type 2 diabetes has become an urban epidemic influenced by neighbourhood environments. However, conventional risk models focusing solely on individual factors fail to account for these neighbourhood influences and often require detailed patient data that may not be available. To address this gap, we developed an integrated approach combining machine learning and causal inference to map type 2 diabetes risk at the neighbourhood level. Using demographic, health, and socioeconomic data from 1,149 Census Tracts (CTs; the neighbourhood unit in this study) in a large metropolitan region, we trained seven machine learning models to identify neighbourhoods with high diabetes prevalence. Although neighbourhood-level diabetes data were available for this study area, our model's high predictive accuracy on external validation data (area under the curve (AUC) = 0.95), particularly from a distinct geographical region, suggests potential utility for predicting diabetes risk in other Canadian regions or elsewhere where such data are unavailable, provided comparable covariates are available and the model is locally retrained and validated using spatially aware procedures. The top models achieved high recall (> 90%) and AUC up to 0.96 on test data, indicating accurate identification of high-risk neighbourhoods with few missed high-risk areas. Survey-derived neighbourhood health indicators, including obesity rate, physical inactivity, and median age were strong predictors of diabetes prevalence. We then applied a Causal Forest approach to estimate conditional average treatment effects (CATE, τ) for selected potentially modifiable factors and summarized the results with the mean $\bar{\tau}$. Higher work stress ($\bar{\tau} = 0.312$) and daily smoking ($\bar{\tau} = 0.155$) were moderately associated with increased risk, whereas better mental health ($\bar{\tau} \approx -1.1$) was protective, highlighting mental health as a priority for further evaluation, especially in neighbourhoods predicted to have high diabetes prevalence. These findings could help identify modifiable neighbourhood-level factors for local prevention efforts and inform equity-oriented planning in diverse urban populations. Prospective or quasi-experimental studies are needed to evaluate intervention effects. Our integrated machine-learning and causal framework lays the groundwork for precision public health, suggesting that modifiable neighbourhood factors may indicate diabetes risk when patient-level data are scarce. Furthermore, the pipeline is conceptually adaptable to other chronic diseases influenced by social and environmental determinants and may inform targeted prevention beyond type 2 diabetes, contingent on disease-specific feature sets and external validation.

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Type 2 diabetes is a growing global health challenge, affecting approximately 422 million people (5.2% of the world population)¹. In Canada, over three million individuals (8.9% of the population) have been diagnosed with diabetes, and 6.1% of adults aged 20 to 79 are prediabetic². Prevalence is rising at approximately 3.3% per year². Although traditional perspectives emphasize biological factors, growing evidence shows that social determinants of health (SDoH) strongly shape diabetes incidence and outcomes^{3–5}. SDoH covers the conditions in which people live and work, including neighbourhood-level resources and stressors. Features such as access to green space, housing stability, and walkability have been linked to better metabolic health outcomes^{6–9}. Similarly, socioeconomic disadvantages, such as differences in income, employment, and education, correlate with a higher diabetes burden^{10,11}. Regional studies in Ontario and Finland further highlight that these social and environmental factors vary across communities, complicating prevention efforts^{12–14}. These complexities suggest the need for analytic methods that capture both the broad influence of environment and the localized nature of risk. Neighbourhood-level approaches can directly link environmental factors to disease dynamics in ways that align with clinical priorities and social service needs¹⁵. In this paper, “neighbourhood” refers to our unit of analysis: the Statistics Canada Census Tract (CT). Neighbourhood-level SDoH data can act as a proxy for missing individual information and can capture broader contextual influences on health behaviours, strengthening the case for integrating fine-grained geographic data into predictive models.

Recent advances in machine learning (ML) have opened new possibilities for health predictive analytics^{16–18}. Researchers have developed ML models for a range of diabetes-related applications, including prognostic models for diabetes complications¹⁹, ensemble and deep learning approaches for predicting diabetes and prediabetes^{20–27}, neural networks to identify mental health comorbidities in diabetes patients^{28,29}, population-level models to explore epidemiological trends³⁰, and survival analyses to estimate time to disease onset^{31,32}. More recently, large language models and generative AI have been explored for diabetes prevention and self-management support^{33,34}. However, deploying such AI systems at a neighbourhood scale remains challenging due to the need for detailed contextual data. Traditional classification and regression methods often provide clearer interpretability of neighbourhood-level factors (e.g., local socioeconomic conditions), especially when data are aggregated or partially observed. Indeed, most ML studies on diabetes rely on individual-level clinical data³⁵, whereas a neighbourhood-oriented perspective can reveal spatial patterns that purely clinical datasets might overlook, particularly in urban areas with sharp socioeconomic gradients³⁶.

Conventional ML is powerful for detecting correlations but cannot tell us if changing a factor will change outcomes³⁷. For example, an algorithm might find that neighbourhoods with more green space tend to have lower diabetes prevalence, but it cannot tell if adding green space would actually *cause* diabetes rates to drop. Causal machine learning (Causal ML) combines principles from statistical inference and ML to address such counterfactual ‘what if’ questions, providing insight into potential outcomes of targeted interventions^{38–40}. Under additional assumptions concerning sufficient adjustment for confounding and temporal ordering of the variables, this approach can guide public health efforts by quantifying how targeting potentially intervenable factors such as walkability or income could influence diabetes prevalence. Although public health agencies collect extensive patient-level diabetes data, these surveillance systems usually focus on confirmed cases and often overlook psychosocial or environmental factors at finer geographic scales. Integrating rich neighbourhood-level variables into predictive models enables ML methods to identify high-risk neighbourhoods and guide localized interventions that account for underlying social and environmental conditions.

In this study, we apply both standard ML and Causal ML methods to predict neighbourhood-level type 2 diabetes prevalence in the Greater Toronto Area (the Toronto Census Metropolitan Area (CMA), population: 6.2 million) and its central municipality, the City of Toronto⁴¹. Toronto’s cultural and ethnic diversity makes it an ideal testbed for examining how multiple determinants interact across neighbourhoods. Our integrated predictive–causal approach is designed to support precision public health (i.e., using data to deliver the right interventions to the right populations) by finding ‘hotspots’ of elevated risk and identifying plausible intervention targets. The insights from this work can help clinicians, policymakers, and urban planners allocate resources and design programs that explicitly account for neighbourhood socioeconomic and environmental conditions. This framework directly links predictive modeling results to real-world impact, illustrating how SDoH profiles vary between neighbourhoods and how data-driven models can guide targeted prevention and improved care.

Results

We evaluated seven machine learning algorithms: Logistic Regression (LR), K-Nearest Neighbours (KNN), Neural Networks (NN), Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF), and Extreme Gradient Boosting (XGBoost). Table 1 compares the performance of these algorithms for detecting high-prevalence neighbourhoods in the Toronto CMA and the City of Toronto, evaluated by accuracy, precision, recall, and F1-score. The SVM and NN models were identified as the top-performing algorithms. The SVM correctly identified all high-prevalence neighbourhoods ($y_i = 1$), achieving 100% recall on the test set. The NN followed closely, with 95% recall on the test set and 78% on the external validation set.

Figure 1 shows that, for the Toronto CMA, the SVM achieved the highest AUC at 0.96 on the test data and 0.95 on external validation, while the NN reached an AUC of 0.94 on the test set and 0.86 on external data. The external validation set for the Toronto CMA and the city of Toronto respectively consists of geographically distinct subsets: Brampton in Peel Region (an area with higher-than-average diabetes prevalence) for the Toronto CMA model and Old Toronto for the City-of-Toronto model. Both subsets include high- and low-prevalence neighbourhoods to ensure a thorough evaluation.

Obesity prevalence, overweight rates, and physical inactivity, as indicated by low active transportation usage, were especially influential in distinguishing neighbourhoods with higher diabetes prevalence, aligning with known links between sedentary lifestyles, excess weight, and metabolic risk^{4,6}. The SVM and NN models appear to capitalize on non-linear relationships among these features by using richer representations. For instance, the

Model	Toronto CMA								City of Toronto							
	Accuracy		Precision		Recall		F1		Accuracy		Precision		Recall		F1	
	T	E	T	E	T	E	T	E	T	E	T	E	T	E	T	E
LR	0.82	0.82	0.51	0.44	0.90	0.67	0.66	0.53	0.81	0.80	0.40	0.40	1.00	0.73	0.54	0.51
KNN	0.86	0.87	0.58	0.55	0.90	0.89	0.70	0.67	0.83	0.84	0.41	0.43	0.93	0.91	0.56	0.59
NN	0.87	0.89	0.59	0.61	0.95	0.78	0.73	0.68	0.82	0.84	0.45	0.45	1.00	0.90	0.62	0.61
SVM	0.86	0.85	0.57	0.52	1.00	0.94	0.72	0.67	0.83	0.77	0.45	0.41	1.00	0.91	0.62	0.62
DT	0.79	0.83	0.45	0.48	0.62	0.83	0.52	0.61	0.84	0.84	0.38	0.45	0.74	0.85	0.50	0.56
RF	0.86	0.82	0.58	0.47	0.86	0.89	0.69	0.62	0.79	0.74	0.37	0.37	0.85	0.86	0.51	0.51
XGBoost	0.75	0.75	0.41	0.38	0.71	0.83	0.52	0.54	0.74	0.78	0.39	0.39	0.63	0.78	0.51	0.51

Table 1. Summary of performance metrics for each model, including accuracy, precision, recall, and F1 score, on test (T) and external validation (E) datasets for the Toronto CMA and the City of Toronto.

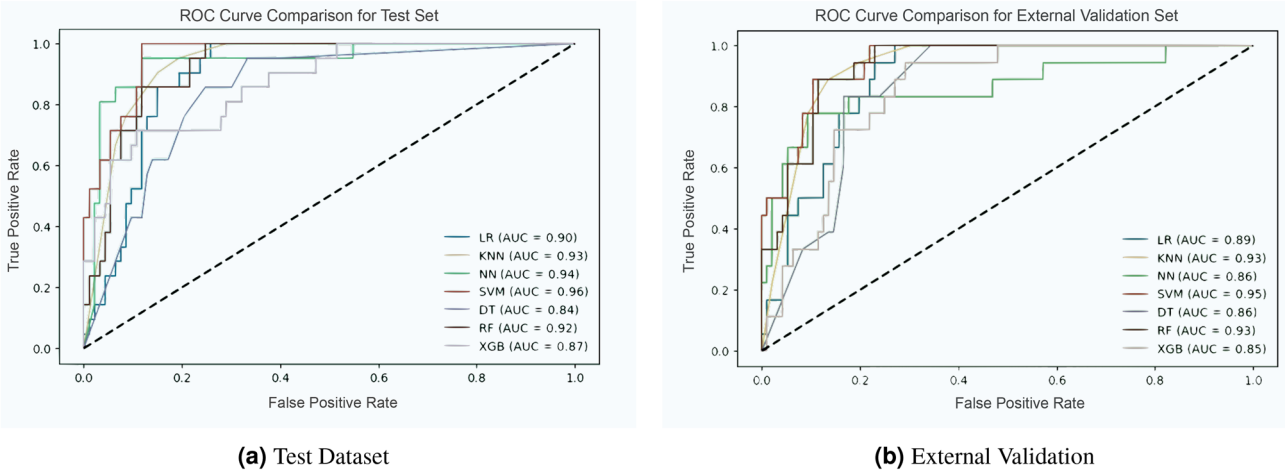


Fig. 1. Receiver operating characteristic (ROC) curve comparison for diabetes prevalence prediction. Each curve shows the true positive rate (TPR) and false positive rate (FPR) across different thresholds. Higher TPRs and lower FPRs shift the curve further away from the diagonal line, which represents random guessing (TPR = FPR). The SVM model outperforms other methods, achieving the highest AUC of 0.96 on the test set and 0.95 on the external validation set.

SVM with a radial basis function kernel can capture complex decision boundaries influenced by socioeconomic and environmental factors, while the NN’s multi-layer architecture learns interactions among variables such as education level, walkability, and obesity. Other algorithms in our set of models (e.g., DT, RF, and XGBoost) can also model non-linearities, but the SVM and NN achieved the highest accuracy and recall on this dataset. For instance, the XGBoost model, despite its popularity, did not outperform the SVM or NN in our study. One possible reason is the relatively limited size of our dataset. Complex ensemble methods like XGBoost can exhibit high variance and may overfit when data are scarce or noisy, which can reduce generalization performance. In such data-constrained settings, the advantages of boosting algorithms are less pronounced, and a well-regularized classifier like the SVM can end up achieving better results.

The SVM’s performance improved markedly after targeted feature engineering. After we log-transformed the skewed median-age variable (hereafter referred to as log median age) and removed highly collinear features, the SVM’s recall on the test set rose from 88% to 100% with no loss of precision. This outcome suggests that normalizing feature distributions and mitigating multicollinearity helped the SVM learn a more effective decision boundary, enabling it to correctly identify all high-prevalence neighbourhoods (achieving 100% sensitivity) without increasing false positives. The subsequent Recursive Feature Elimination with Cross-Validation (RFECV) procedure identified five leading predictors that were consistently retained by the SVM, NN, RF, KNN, and LR models: obesity rate, overweight rate, active transportation rate, rate of physically active population, and log-transformed median age. These factors are closely tied to local lifestyle and demographic conditions, which highlight the value of combining individual health-behaviour indicators with environmental context when modeling diabetes risk.

Geographical Applicability and Model Scalability

The Toronto CMA encompasses a large and diverse region, whereas the City of Toronto represents the dense urban core with somewhat more homogeneous demographics and narrower socioeconomic variability. We

observed that applying the SVM model (trained on the CMA) to data from the City of Toronto led to a drop in precision (from 57% to 45% on the test set, and from 52% to 41% on external validation), while recall remained 100% on the test set and dipped only slightly (from 94% to 91%) on external validation (Table 1). One likely explanation is that the City of Toronto has less variability in certain predictors, which caused the classifier to flag a larger share of neighbourhoods as high-prevalence (i.e., increasing false positives). In particular, in the City of Toronto, about 20% of the neighbourhoods have high prevalence, so a precision of 45% still represents more than double the base rate. From a public health perspective, we therefore prioritized recall (i.e., sensitivity) over precision because reliably identifying every high-prevalence neighbourhood, even at the expense of some false alarms, is preferable to missing neighbourhoods that need intervention.

Causal Forest analysis of diabetes risk factors

We conducted a Causal Forest analysis to explore heterogeneous associations of selected neighbourhood factors with diabetes prevalence. Three sets of features were defined to explore different dynamics: (1) a *base* set consisting of the SVM’s top predictive features (i.e., log median age, obesity rate, overweight rate, active transportation rate, and active population rate), (2) an *extended* set that added key psychosocial factors to the base (i.e., mental health score, visible minority rate, work stress score, and daily smokers rate), and (3) a *full* feature set incorporating a broader range of attributes (Table 2). The extended set was designed to introduce modifiable social and behavioural variables (e.g., stress and smoking) while keeping model complexity in check.

We used Causal Forest to estimate the mean Conditional Average Treatment Effect (CATE), denoted by $\bar{\tau} = E_X[\tau(X)]$, for selected features to explore the heterogeneous relationships between predictors and diabetes outcomes. $\bar{\tau}$ represents the average of the conditional effects (slopes) of each treatment on the modeled probability of Diabetes = 1, averaged over the observed covariate profiles of neighbourhoods. The outcome was modeled on the percent-prevalence scale, so $\bar{\tau}$ values are expressed in percentage-point (pp) changes in diabetes prevalence per one-unit change in the treatment. For rate variables, the unit step corresponds to a +1 percentage-point increase; for 1–5 score variables (mental health score, work stress score, and food insecurity score), it corresponds to a +1 scale-point increase. Because a one-point change on a 1–5 scale represents a much larger shift than a one-percentage-point change in a rate, direct comparisons of $\bar{\tau}$ magnitudes across these variable types should be made with caution. Note that $\bar{\tau}$ denotes an effect (a slope) rather than a probability; large local slopes (e.g., $|\bar{\tau}| > 1$) are possible even though modeled probabilities remain bounded within [0,1].

Some candidate variables were not treated as direct ‘interventions’ in this analysis. For example, we did not assign treatments for log median age, obesity, overweight, visible minority rate, or recent immigrant rate, because these factors are difficult to manipulate in isolation or are inherently tied to other variables. Reducing obesity at a population level, for instance, requires multifaceted programmes involving policy changes, dietary interventions, and increased physical activity, and recent immigrant status is closely linked to income and education. We therefore focused on feasible intervention-related features. When considering a specific variable in the feature set as the treatment, the other variables enter the analysis as confounders/effect modifiers.

For the base feature set, the Causal Forest estimated almost no direct effect from increasing active transportation ($\bar{\tau} = 0.003$) or the active population rate ($\bar{\tau} = -0.011$). This outcome does not mean that promoting physical activity has no value, only that within our dataset and model these variables did not show a strong mean CATE estimate on diabetes prevalence. With the extended feature set, which added mental health, visible minority status, work stress, and smoking, the analysis revealed a large negative mean CATE for mental

Feature set	Treatment	$\bar{\tau}$
SVM set (Log median age, obese rate, overweight rate, active transportation rate, rate of active population)	Active transportation rate	0.003
	Rate of active population	– 0.011
Extended set (SVM features + mental health score, visible minority rate, work stress score, daily smokers rate)	Active transportation rate	0.004
	Rate of active population	– 0.001
	Mental health score	– 1.113
	Work stress score	– 0.021
	Daily smokers rate	0.155
Full set (log median age, obese rate, overweight rate, rented rate, residential instability index, material deprivation, ethnic concentration index, food insecurity score, visible minority rate, work stress score, mental health score, regular alcohol drinkers rate, daily smokers rate, rate of active population, active transportation rate, high education rate, average income, unemployment rate, recent immigration rate)	Food insecurity score	– 0.039
	Work stress score	0.312
	Mental health score	– 1.045
	Regular alcohol drinkers rate	– 0.005
	Rate of daily smokers	0.003
	Rate of active population	– 0.017
	Active transportation rate	– 0.014
	High education rate	0.001
	Average income	0.000
	Unemployment rate	– 0.001

Table 2. Summary of Causal Forest results for each feature set (SVM base, extended, and full), reporting the estimated $\bar{\tau}$ for selected treatments in neighbourhood-level diabetes prevalence.

health, $\bar{\tau} = -1.113$, and a moderate positive mean CATE for daily smoking, $\bar{\tau} = 0.155$, consistent with the known metabolic risks of smoking.

In the model incorporating the full feature set, work stress showed a moderate positive mean CATE on diabetes prevalence, $\bar{\tau} = 0.312$, whereas mental health remained strongly protective, $\bar{\tau} = -1.045$. Food insecurity displayed a slight negative coefficient, $\bar{\tau} = -0.039$, which may reflect a complex mix of household factors or simply noise given the small magnitude. Alcohol use, unemployment rate, and recent immigration rate produced essentially negligible mean CATEs in our Causal Forest results, suggesting that other psychosocial and environmental features play larger roles in determining neighbourhood-level diabetes vulnerability.

Discussion

Neighbourhood-level data provided a clear advantage in identifying neighbourhoods with high diabetes prevalence, beyond what individual-level or highly aggregated data alone can offer. When combined with physical activity indicators, factors such as obesity, overweight rates, and age structure were identified as important predictors of local risk, showing how the built environment and lifestyle factors shape disease patterns^{42,43}. These findings are consistent with previous research linking excess body weight and sedentary behaviour to cardiometabolic conditions in urban settings⁴. However, many health agencies lack standardized neighbourhood-level data on outcomes and risk factors, making it difficult to monitor socioeconomic disparities in diverse populations⁴⁴. Systematically incorporating local characteristics could refine resource allocation by identifying areas where interventions are most urgently needed⁴⁵.

Assessing model reliability and sensitivity

To evaluate model reliability, we measured the sensitivity of the SVM to each predictor through permutation tests⁴⁶. For a given performance metric $M(f)$, the values of feature x_j were randomly permuted across the dataset, breaking its link to the outcome, and the resulting drop in performance was recorded. Feature importance was then defined as

$$S(x_j) = M(f(x)) - M(f(x^{\pi_j})),$$

where $M(f(x))$ is the original score and $f(x^{\pi_j})$ is the model after permuting feature j . A larger $S(x_j)$ therefore indicates a stronger contribution of that feature⁴⁷. Figure 2a and b show that permuting Log Median Age produced the greatest reduction in recall and F1, making it the most influential variable for identifying high-prevalence neighbourhoods. Obesity, Overweight, and Physical Activity followed in importance, a pattern consistent with evidence that active living reduces metabolic risk in urban settings⁴⁸.

We complemented this global analysis with SHapley Additive exPlanations (SHAP), which allocate the difference between an individual prediction and a baseline prediction to each feature i ⁴⁹:

$$\phi_i(x) = \mathbb{E}_{S \subseteq \mathcal{F} \setminus \{i\}} [f(S \cup \{i\}, x) - f(S, x)].$$

The beeswarm plots in Fig. 2a (i.e., internal test set) and Fig. 2b (i.e., external validation set) display these local contributions. Neighbourhoods with higher Log Median Age or Obesity shift SHAP values to the right, pushing predictions toward the 'high-prevalence' class, whereas greater use of active transportation shifts points to the left and signals lower risk. The accompanying bar charts confirm that the features with the largest SHAP magnitudes also produce the steepest declines in recall and F1 when permuted. The agreement between SHAP and permutation sensitivity indicates that age structure, obesity, and physical-activity measures are strong predictive markers of high-prevalence neighbourhoods; these associations can inform targeting and motivate interventions on upstream determinants that shape obesity and activity environments, consistent with prior work^{50,51}.

Urban lifestyle interventions

Our results indicate that urban design elements that facilitate physical activity are associated with neighbourhood-level diabetes risk and can shape behaviours linked to that risk. Prior work shows that the built environment shapes daily movement patterns and metabolic health⁵². Although the mean CATE for active transportation, $\bar{\tau}$, was close to zero in our analysis, this finding reflects limitations of the current dataset rather than the irrelevance of the intervention. In practice, redesigning neighbourhoods, for example by adding bike lanes or continuous sidewalks, can promote more active routines and may lower diabetes prevalence in high-risk areas. Well-planned infrastructure is widely recognized as a catalyst for healthier behaviours and can, over time, reduce the metabolic burden on urban communities.

Evidence from other cities supports this interpretation. neighbourhoods designed with pedestrians and cyclists in mind tend to have lower diabetes prevalence⁵³. In Toronto, for instance, debates over removing bike lanes have raised concerns about negative impacts on residents' daily activity and well-being⁵⁴. Likewise, experiences in cities like Copenhagen and Melbourne show that urban designs promoting active transportation are associated with improved metabolic health indicators⁵⁵. Importantly, urban lifestyle interventions carry co-benefits beyond diabetes prevention: features that support walking and cycling also contribute to cleaner air and stronger social cohesion in communities⁵⁶. Policymakers can use these insights to align urban planning with public health goals, clarifying how specific design choices in the built environment may reinforce healthier lifestyles.

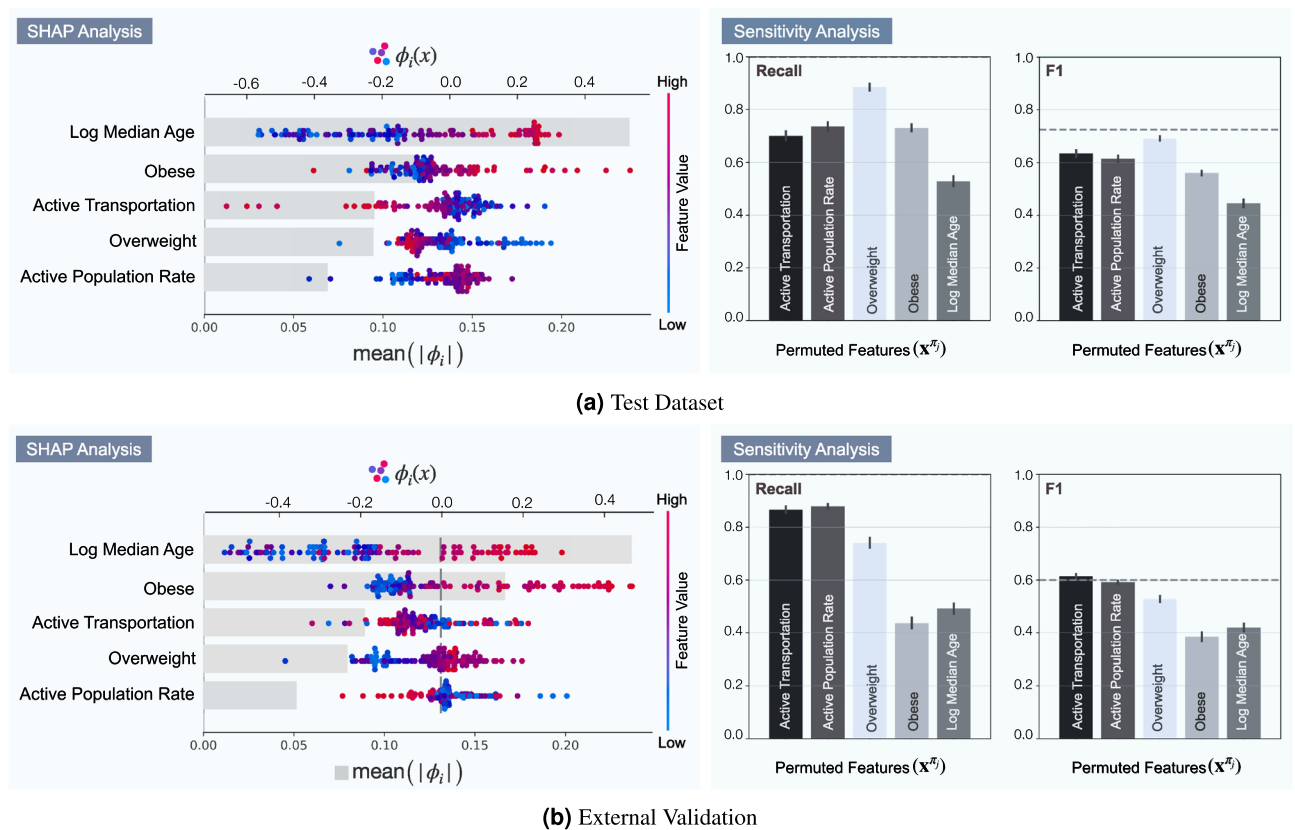


Fig. 2. Integrated SHAP and sensitivity analysis of SVM interpretability and robustness. **(a)** Test dataset. The upper beeswarm displays local Shapley values $\phi_i(x)$ for every feature, coloured from pink to blue to denote low and high feature values. Points to the right of zero increase the predicted probability of high prevalence, whereas points to the left decrease it. The lower bar chart reports global importance through mean absolute Shapley values $\mathbb{E}[|\phi_i|]$. The adjacent bars show the drop in Recall and F1 when feature j is permuted, x^{π_j} , thereby measuring sensitivity. **(b)** External-validation dataset. The same structure confirms that Log Median Age, Obesity, Overweight, Active Population Rate, and Active Transportation remain the dominant predictors, and that the permutation analysis yields comparable robustness across data sources.

Beyond individual design elements, our results align with integrated planning approaches such as the ‘15-minute city’, which aim to place daily needs within a short walk or cycle. By co-locating housing, essential services, parks, and transit, these frameworks can increase routine physical activity, reduce car reliance, and improve access to healthy food and green space. If implemented with equity safeguards to avoid displacement, such policies could help lower neighbourhood diabetes burden and narrow disparities, with potential co-benefits for mental health that our analysis identified as protective. We recommend that implementations be accompanied by spatially aware external validation and quasi-experimental evaluation to test whether changes in walkability, food access, and greenness translate into measurable reductions in diabetes prevalence.

Ethnocultural influences on diabetes

We used an identical set of five features for the Toronto CMA and the City of Toronto ML models to ensure comparability on the two geographic scales. To achieve this, we removed two variables, mental health score and work stress score, from the City of Toronto model due to multicollinearity. To explore the impact of these psychosocial factors on ML prediction, we conducted an extended analysis on the Toronto CMA data that explicitly included mental health score, work stress score, log median age, obesity, overweight, active transportation, and active population rate (seven features). Adding the mental health and work stress variables led to slight performance changes: approximately a 2% gain in accuracy, a 6% gain in precision, and a 1% gain in F1, accompanied by a 3% drop in recall. This outcome aligns with earlier observations that expanding a model’s feature set can boost accuracy and precision at the cost of a small decrease in sensitivity^{57,58}. The trade-off arises because additional predictors introduce variability, making the model more confident (and precise) in classifying many positive cases, but causing it to miss a few borderline cases it previously captured. Nonetheless, incorporating mental health and work stress measures enhanced the model’s ability to capture local sociocultural factors. In a diverse region like the Toronto CMA, these variables noticeably affect diabetes risk, thus their inclusion helps the model better reflect real-world neighbourhood differences.

One notable insight from our analysis is the association between neighbourhood ethnic composition and diabetes. The *visible minority rate* was a key predictor in our models. Furthermore, the line graph in Fig. 3a indicates an inverse relationship: neighbourhoods with higher proportions of visible minority residents tended to have lower average diabetes prevalence. This pattern is consistent with some earlier findings on how neighbourhood racial/ethnic composition and segregation relate to diabetes disparities^{59,60}. It may suggest that neighbourhoods with many recent immigrants or minority residents maintain certain protective cultural practices (e.g., traditional diets or strong social networks) that help mitigate diabetes risk⁶¹. The well-documented 'healthy immigrant effect' may also contribute to this pattern. Recent immigrants often arrive in relatively good health because of selection mechanisms, and this advantage can translate into lower chronic disease rates in the short term⁶². In Toronto, where more than 50 % of residents are immigrants⁶³, these findings show how demographic context can shape the local diabetes burden.

A similar protective gradient emerged for perceived mental health. We observed a clear inverse association between neighbourhood mental health status and diabetes burden. Neighbourhoods with *higher* mean mental health scores, indicating better perceived well-being, exhibited *lower* mean diabetes prevalence, as illustrated in Fig. 3b. This pattern is consistent with extensive epidemiological evidence showing that psychological distress and depression elevate the risk of incident type 2 diabetes and worsen glycemic control and insulin sensitivity, contributing to diabetes development^{64–68}. At the physiological level, chronic psychosocial stress activates the hypothalamic–pituitary–adrenal axis, elevates cortisol, and promotes systemic insulin resistance, linking poor mental health to metabolic dysfunction⁶⁹. Conversely, the demands of diabetes self-management can negatively affect mental well-being, creating a bidirectional feedback loop between the two conditions³⁰.

It is important to acknowledge that the Canadian Community Health Survey (CCHS) item is a single, self-rated indicator. Although validated against structured diagnostic interviews⁷⁰, self-report instruments can vary in meaning across cultural groups and may underrepresent individuals experiencing severe psychological distress, especially when survey non-response is high^{71,72}. Such biases typically weaken rather than exaggerate associations, so the true mental-health gradient in diabetes prevalence may be even steeper than observed. Our supplementary analyses (Supplementary Figs. 1 and 2) further illustrate how mental health and work stress influenced the Toronto CMA models under the extended and full feature sets. Taken together, these results suggest that strategies addressing both psychological well-being and metabolic health could yield better outcomes than focusing on either in isolation.

Causal Forest insights on diabetes risk

Our Causal Forest analysis highlighted factors with larger estimated mean CATE in this cross-sectional setting. *Work stress* showed a moderate positive mean CATE $\bar{\tau} = 0.312$ in the full model, consistent with evidence that psychosocial stress impairs metabolic function⁷³. *Daily smoking* showed a smaller positive mean CATE, with $\bar{\tau} = 0.155$, aligning with literature that connects smoking to inflammation and insulin resistance. In contrast, *mental health* exhibited a large negative mean CATE. We estimated $\bar{\tau} \approx -1.1$ for the mental health score across both the extended and full feature sets, implying that better average well-being in a neighbourhood could noticeably lower its diabetes prevalence. Nevertheless, prospective or quasi-experimental evaluation is needed to assess effects.

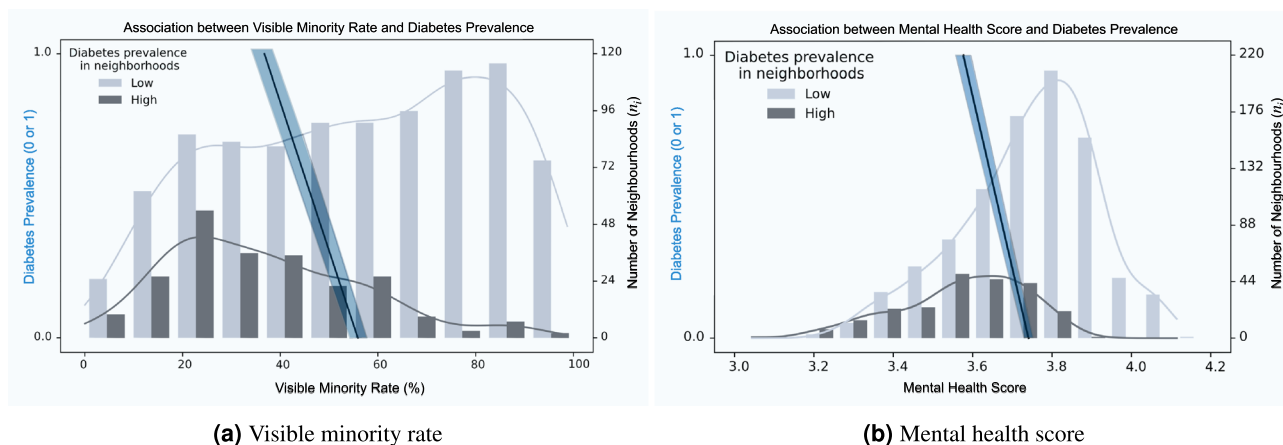


Fig. 3. Visible Minority Rate and Mental Health Score in relation to diabetes prevalence. (a) shows a grouped bar chart of binned visible-minority rates in neighbourhoods classified by diabetes prevalence (low: $y_i = 0$, high: $y_i = 1$). Bars (right y-axis) indicate the number of neighbourhoods n_i in each bin, and smoothed density curves overlay the bars to highlight where each class is concentrated. The fitted line visualizes the mean visible minority rate for each diabetes class, with diabetes prevalence class on the y-axis and the visible minority rate on the x-axis. For each diabetes class, the mean visible minority rate is calculated across the set of census tracts (CTs) labeled with diabetes class i as $\hat{x}_i = \frac{1}{n_i} \sum_{\ell \in C_i} x_\ell$, where C_i denotes the set of CTs with diabetes class i , $n_i = |C_i|$ is the number of such CTs, and x_ℓ is the visible minority rate of CT ℓ . (b) applies the same approach to mental health scores, where x_ℓ denotes the mental health score of tract ℓ .

The Causal Forest findings are supported by our model interpretability analysis. Although active transportation did not exhibit a notable effect ($\bar{\tau} \approx 0$) in the Causal Forest results, it remained an important predictor in the ML models' SHAP rankings. This discrepancy suggests indirect or interaction pathways. For example, neighbourhoods that promote walking may also provide other health-supporting amenities. Age structure, captured by `log median age`, consistently ranked among the top predictors and was associated with higher diabetes risk, a pattern expected for an age-related disease.

Obesity repeatedly appeared as a key driver in the SHAP summaries. Our combined results further suggest an interaction between work stress and weight status. The adverse effect of stress was stronger in neighbourhoods with high rates of overweight residents than in those with high obesity rates. A possible explanation is that individuals who are overweight but not yet obese may be more susceptible to stress-induced metabolic changes, with stress acting as a tipping point⁷⁴. Such context-specific interactions would be missed in a single-factor analysis.

Together, causal inference and SHAP analysis provide a more comprehensive view of the pathways that link psychosocial and lifestyle factors to diabetes risk⁷⁵. Supplementary Figs. 3 and 4 present additional results from the base, extended, and full feature sets, and the combined evidence clarifies the roles of psychosocial and demographic attributes, such as mental health, stress management, and age structure, in shaping neighbourhood-level diabetes burden.

Limitations

Although we included a broad range of socioeconomic and behavioural variables, several environmental factors known to influence health were not directly measured. For instance, the dataset lacked explicit information on green space availability and air-pollution levels⁷⁶. We relied on proxies such as active-transportation metrics and material-deprivation indices to approximate these exposures, but direct measurements would likely improve model performance. The Causal Forest residual term, $\varepsilon = y - \hat{y}$, may therefore capture unobserved environmental influences or latent confounders. Working at the census-tract scale can also obscure finer social and mobility patterns that shape health outcomes, a limitation commonly referred to as the *modifiable areal unit problem* (MAUP)⁷⁷. We did not explicitly model spatial dependence among census tracts (for example, spatial lag or spatial error structures), and we did not apply spatial cross-validation. If residual spatial autocorrelation is present, random cross-validation can yield optimistic performance estimates and anti-conservative uncertainty. Future work can incorporate spatial diagnostics and spatially explicit approaches (e.g., spatial filtering or models with spatial random effects) together with block or cluster cross-validation to assess robustness.

Several neighbourhood exposures were not directly measured at the census-tract level, including diet quality and the retail food environment (e.g., supermarket density, healthy food availability/price) and greenness (e.g., NDVI, park proximity/acreage). We used food insecurity as a proxy for diet constraints and active transportation/material deprivation as partial proxies for aspects of the built environment, but residual confounding may remain. Because greenness and food access plausibly correlate with both behaviours (physical activity, diet) and diabetes prevalence, their omission could bias effect estimates (e.g., attenuating or, depending on correlation structure, inflating associations attributed to active transportation). Future work should link tract-level remote-sensing and administrative layers (NDVI, parks, walkability) and retail food-environment data, adopt spatial cross-validation with sensitivity and negative-control analyses, and, where feasible, leverage quasi-experimental settings (e.g., new parks, bike-lane installations, or grocery-store openings) to reduce omitted-variable bias and strengthen causal inference.

Many predictors originated from self-reported survey data, introducing potential biases such as recall error and variation in how questions are interpreted across cultural groups⁷⁸. We cross-validated with additional data sources whenever possible. However, more objective clinical measures such as biometric indicators would enhance model fidelity. The cross-sectional design also limits causal interpretation, so the Causal Forest estimates should be interpreted with caution. Interpretation of these as causal effects would require assuming no unmeasured confounding, and while causal inferences based on cross-sectional data are possible, this requires even stronger assumptions than other observational study designs⁷⁹. Longitudinal or quasi-experimental approaches, such as natural experiments, would provide clearer evidence of how changes in social and environmental conditions influence diabetes prevalence over time^{80–83}. We also assume no spillover between tracts (SUTVA) and adequate treatment–control overlap; violations (e.g., spatial spillovers or poor overlap) would bias the CATE estimates.

Moreover, our external validation was limited to the Toronto CMA and the City of Toronto, which may constrain generalizability⁸⁴. Applying this framework to other urban regions and to rural settings would strengthen confidence in its broader applicability. Data limitations also prevented inclusion of certain relevant factors, such as neighbourhood health care access or culturally specific dietary behaviours, that can influence diabetes risk and model completeness. In addition, some population subgroups may be underrepresented in the data, potentially introducing prediction bias⁸⁵. Future work should incorporate richer information on health status, income and housing, in line with recommendations for comprehensive and equitable ML practice in health care⁸⁶. Also, generalization to new cities will require comparable covariates, local retraining, and spatially aware external validation.

Despite these limitations, our study demonstrates the unique value of integrating ML prediction with causal inference in public health analytics. This combined approach offers a novel blueprint for precision public health⁸⁷: it clarifies how modifiable neighbourhood factors relate to disease risk and helps ensure that the right interventions reach the right communities. The same methodology could be adapted to other chronic diseases influenced by social and environmental determinants, potentially guiding targeted prevention efforts well beyond type 2 diabetes.

Methods

Figure 4 provides an overview of our machine learning and causal inference pipeline for identifying neighbourhoods with high type 2 diabetes prevalence in the Toronto CMA. The process encompasses data collection, feature engineering, predictive modeling with seven ML algorithms, and interpretative analyses supported by sensitivity checks, SHAP analysis, and causal ML to explore counterfactual outcomes. We formalize our dataset as $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where each neighbourhood i is represented by a feature vector $x_i \in \mathbb{R}^m$ (with m features) and a binary label y_i indicating whether the neighbourhood falls in the top quintile of diabetes prevalence (10.5% to 23.8%). We train a classification model M to predict high-prevalence status, such that $\hat{y}_i = M(x_i)$ should correctly flag neighbourhoods at elevated risk.

Dataset overview and feature engineering

Dataset overview

Census tracts (CTs) defined by Statistics Canada serve as proxies for neighbourhoods with relatively homogeneous socioeconomic and living conditions⁸⁸. We selected predictor variables based on prior studies linking health, socioeconomic, lifestyle, and marginalization indicators to diabetes outcomes^{4,6,8,9,11}. Neighbourhood-level diabetes prevalence data came from the 2022 Canadian Community Health Survey (CCHS)⁸⁹, which samples Canadians aged 12 and older across 121 health regions and covers about 98% of the population. The remaining 2% live in remote locations such as reserves or Crown Lands and are excluded because of logistical constraints. From the CCHS we identified 1,149 CTs in the Toronto CMA, of which 230 (20.0%) met the definition of high-prevalence. We classified a tract as 'high-prevalence' if its diabetes rate exceeded the 80th percentile threshold of 10.5%. Figure 5a shows the spatial pattern of these tracts, and Fig. 5b presents the full distribution of prevalence values up to the maximum of 23.8%. This binary categorization, contrasting tracts in the top quintile with all others, allowed us to concentrate on areas at distinctly elevated risk.

Data exploration and feature engineering

Neighbourhood predictors were drawn from three main sources: (i) health and lifestyle measures from the 2022 CCHS, (ii) demographic and socioeconomic variables from Environics Analytics' DemoStats dataset⁹⁰, and (iii) marginalization indices from the 2021 Ontario Marginalization Index (ON-Marg) compiled by the MAP Centre for Urban Health Solutions⁹¹. DemoStats provides more than 750 indicators, derived from census counts, economic data, and immigration information, at the CT level⁹². ON-Marg supplies dimension-specific indices of neighbourhood marginalization, such as economic disadvantage and residential instability, together with quintile groupings⁹³. We accessed these sources through the SimplyAnalytics platform⁹⁴ and Public Health Ontario⁹⁵, which yielded an initial pool of 26 candidate features summarized in Table 3. Our pipeline includes a missing data analysis step; however, this dataset contained no missing values across the census tracts, so that step was skipped and no imputation or case-wise deletion was required.

We applied standard preprocessing steps to prepare the features. Categorical variables were one-hot encoded. We created interaction terms to capture higher-order socioeconomic patterns. For instance, we combined the material deprivation and residential instability quintiles into a single 'instability deprivation' indicator that represents neighbourhoods facing both economic and housing challenges⁹¹. We applied log transformations to skewed variables (e.g., median age) to reduce skewness and checked for multicollinearity, excluding any feature with a Pearson correlation $|r| \geq 0.7$ with another feature.

The dataset was split into a training set (80% of neighbourhoods), a test set (10%), and an external validation set (10%). Feature scaling (i.e., normalization) was fitted on the training data and then applied to the test and validation data. Given the 20% prevalence of the positive class, we balanced the training set using RUS, reducing 919 training examples to 340 while leaving the class distribution unchanged in the test and validation sets. For external validation, we chose two geographically distinct areas: Brampton in Peel Region for the Toronto-CMA model and Old Toronto (along Lake Ontario) for the City-of-Toronto model. Peel Region, which includes

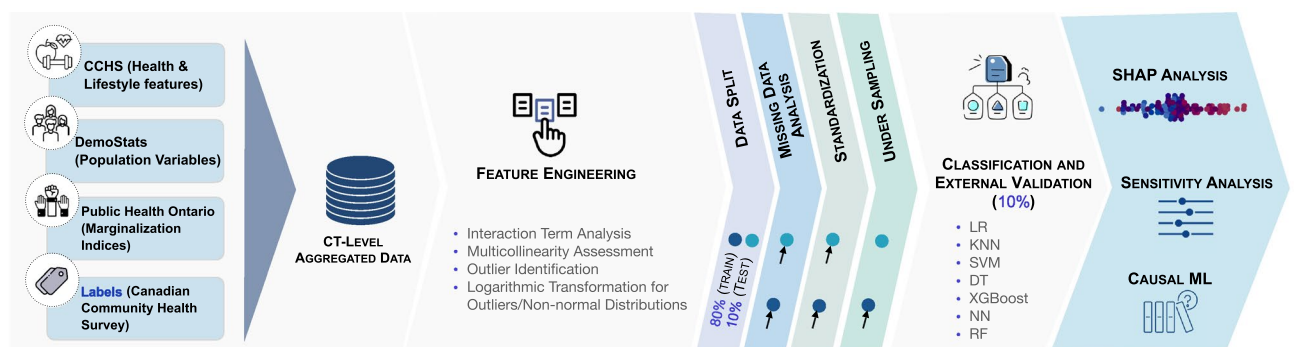


Fig. 4. Overview of the employed ML and causal ML pipeline. The workflow combines data from multiple sources, addresses interactions and class imbalance, then trains seven ML algorithms (LR, KNN, SVM, DT, XGBoost, NN, RF). SHAP, sensitivity analysis, and Causal ML clarify model outputs and suggest potential neighbourhood interventions. The pipeline includes a general missing data analysis step; in this study the dataset contained no missing values, so this step was skipped.

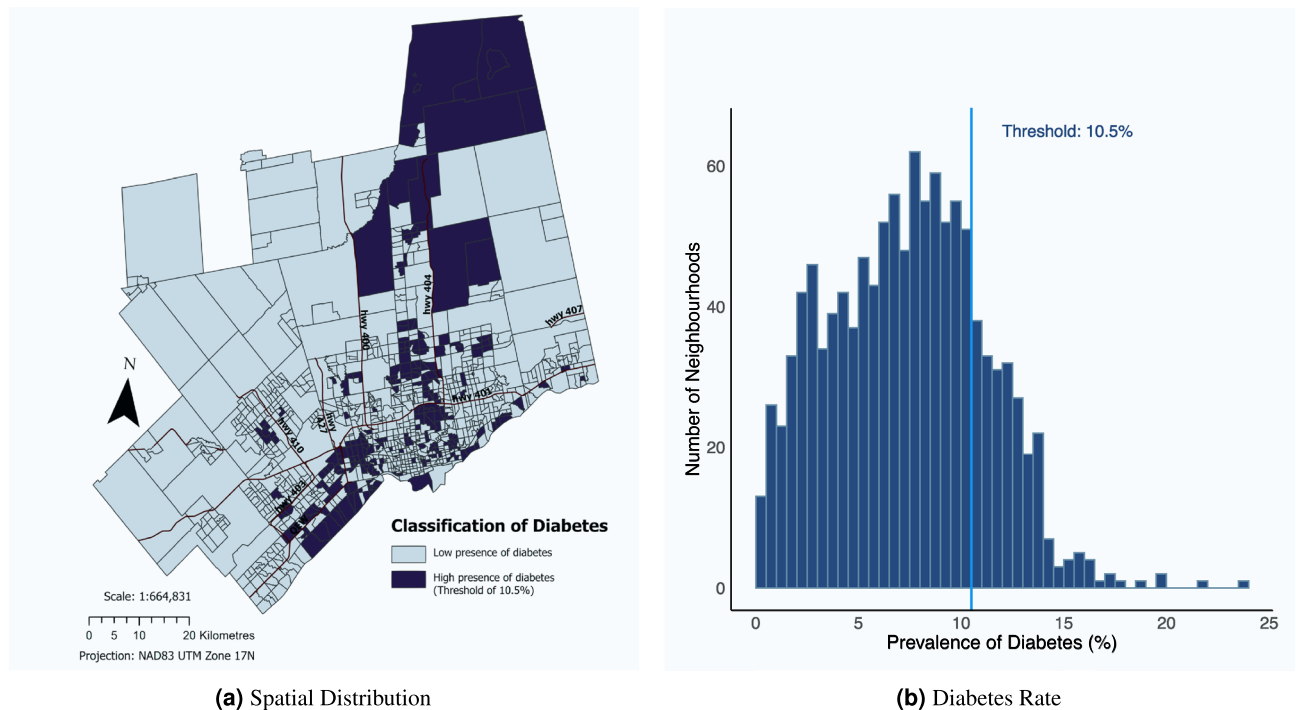


Fig. 5. Geographic distribution and prevalence rates of type 2 diabetes in the Toronto CMA. **(a)** Map of neighbourhoods classified as high-prevalence (10.5%–23.8%) or low-prevalence (lower than 10.5%). The map was generated using ArcGIS Pro 3.2 (Esri, Redlands, CA; <https://www.esri.com/en-us/arcgis/products/arcgis-pro/overview>). **(b)** Histogram showing the distribution of Type 2 diabetes prevalence rates across 1,149 neighbourhoods in the region.

Brampton, reports a higher diabetes prevalence than the Ontario average⁹⁶, making it a stringent test case. Each external subset contained both high- and low-prevalence neighbourhoods to ensure a thorough evaluation. We used five-fold cross-validation during training for hyperparameter tuning and feature selection, which helped improve model stability and prevent overfitting.

Handling class imbalance and probability calibration

In this study, the positive class (i.e., $y_i = 1$: high-prevalence neighbourhoods) accounted for roughly 20% of the sample, producing a class imbalance that can bias ML models toward the majority class. To address this, we applied Random Under-Sampling (RUS) to the training data: we retained all minority-class examples and down-sampled the majority class to achieve a desired ratio. Let N_{maj} and N_{min} denote the numbers of majority and minority observations, respectively. We drew a subset $S \subseteq N_{\text{maj}}$ with $|S| = r N_{\text{min}}$ for a chosen ratio r , then trained the model on $S \cup N_{\text{min}}$. This strategy preserved authentic high-prevalence neighbourhoods rather than synthetically generating new ones. We chose RUS over oversampling methods, such as the synthetic minority oversampling technique (SMOTE)⁹⁷, which creates synthetic data points that could distort complex socioeconomic feature distributions. For example, SMOTE interpolates between minority instances as

$$x_{\text{new}} = x_i + \lambda(x_j - x_i), \quad \lambda \sim U(0, 1),$$

where λ is a random interpolation weight drawn from the continuous uniform distribution $U(0, 1)$ that determines how far along the segment from x_i to x_j the synthetic point lies. Although SMOTE-Nominal Continuous (SMOTE-NC) adapts this approach for mixed data, synthetic points may misrepresent certain categorical or survey-derived characteristics essential to our local context. RUS preserves authentic minority samples, aligning with our goal of maximizing recall for higher-risk neighbourhoods⁹⁸.

We also explored probability calibration with methods such as Platt scaling or isotonic regression. Platt scaling fits a logistic function,

$$P(y = 1 | s) = \frac{1}{1 + \exp(As + B)},$$

where s is the uncalibrated score, and (A, B) are fitted parameters. Isotonic regression learns a non-decreasing piecewise-constant function to map raw scores to probabilities. While these calibration techniques can improve the interpretability of probability outputs, they add model complexity and require additional data to fit reliably. In our case, with a relatively limited dataset, we prioritized identifying high-prevalence neighbourhoods (i.e.,

Feature	Description
Canadian Community Health Survey	
Mental health score	Perceived mental health: Higher = better
Mental health binary*	0 = poor, 1 = good (avg score cutoff)
Regular alcohol drinkers rate	% regular alcohol drinkers
Daily smokers rate	% daily smokers
Food insecurity score	Food insecurity: Higher = more
Food insecurity binary*	0 = low, 1 = high (avg score cutoff)
Work stress score	Perceived work stress: Higher = more
Work stress binary*	0 = low, 1 = high (avg score cutoff)
Rate of active population	% physically active (18+)
Active transportation rate	% using active transport (18+)
Obese rate	% obese population
Overweight rate	% overweight population
DemoStats	
High education rate	% with diploma/degree (25-64)
Average income	Avg household income
Unemployment rate	% unemployed (15+)
Median age	Median population age
Recent immigrant rate	% immigrants (2017-2022)
Visible minority rate**	% visible minorities
Visible minority binary*	0 = <25%, 1 = >25%
Rented rate	% rented dwellings
Public Health Ontario	
Ethnic concentration index	% recent immigrants & visible minorities
Ethnic concentration quintiles	Ethnic concentration quintiles
Residential instability index	Residential density & family structure
Residential instability quintiles	Residential instability quintiles
Material deprivation index	Access to basic needs
Material deprivation quintiles	Material deprivation quintiles

Table 3. Overview of the 26 study variables sourced from the Canadian Community Health Survey (CCHS), DemoStats, and Public Health Ontario, representing population health, demographics, and socio-economic status. *Feature calculated by the authors using the raw feature. **The visible minority population consists mainly of the following groups: South Asian, Chinese, Black, Filipino, Arab, Latin American, Southeast Asian, West Asian, Korean and Japanese (Statistics Canada-Visible Minority of a person, 2021).

recall and F1) over perfectly calibrated probabilities. Our primary objective was classification performance, flagging as many true high-risk neighbourhoods as possible, so we elected not to apply post-hoc calibration, as it was not essential for the intervention-focused goals of this study.

Careful curation of neighbourhood variables improved model performance and identified potential targets for interventions such as material deprivation and residential instability, which can guide infrastructure upgrades and resource allocation for diabetes prevention. This same feature-driven framework can be adapted to other chronic diseases influenced by environment and behaviour, including hypertension.

Predictive modeling

A hypothesis class $\mathcal{H}$ with seven ML architectures was chosen to reflect varied learning capacities and interpretability: LR, KNN, NN, SVM, DT, RF, and XGBoost. These algorithms differ in complexity, feature handling, and parameterization, allowing an in-depth examination of neighbourhood-level heterogeneity. Each model learns a decision rule $h \in \mathcal{H}$ to minimize a loss function on the training set, with hyperparameters tuned by five-fold cross-validation.

Model performance was evaluated in terms of accuracy, precision, recall, F1-score, and area under the ROC curve (AUC) to provide a comprehensive assessment. Since our primary goal was to correctly identify high-prevalence neighbourhoods, we paid particular attention to recall (i.e., sensitivity). We optimized hyperparameters for each algorithm using grid search with cross-validation. Cross-validation folds were stratified by class and assigned at random across census tracts; we did not enforce spatial blocking. Moreover, for models where feature selection could simplify the model without sacrificing performance (i.e., LR, NN, DT, RF, and XGBoost), we employed RFECV to prune away redundant features. We visualized model outcomes using confusion matrices and ROC curves on both training and test sets to check for any overfitting and to compare performance across models. After identifying the strongest performers, we further evaluated those models on the independent external validation sets to assess generalizability. To interpret the predictions, we applied SHAP

Model	# of features	Optimal set of features	Model parameters
LR	5	Obese rate, Overweight rate, Active transportation rate, Rate of active population, Log median age	$C^{[1]}$: 0.3, <i>penalty</i> : l1, <i>solver</i> : liblinear
KNN	5 ^[2]	Obese rate, Overweight rate, Active transportation rate, Rate of active population, Log median age	Optimal <i>k</i> (using Elbow method): 6, <i>algorithm</i> : auto, <i>metric</i> : cosine and <i>weights</i> : uniform
NN ^[3]	7	Obese rate, Overweight rate, Active transportation rate, Rate of active population, Log median age, Visible minority rate, Daily smokers rate	<i>activation</i> : tanh, <i>hidden_layer_sizes</i> : (128, 64), <i>solver</i> : lbfgs
SVM	5 ^[2]	Obese rate, Overweight rate, Active transportation rate, Rate of active population, Log median age	$C^{[1]}$: 10, <i>gamma</i> : 0.1, and <i>kernel</i> : rbf ^[4]
DT	4	Obese rate, Overweight rate, and Log median age, Visible minority rate	<i>max_depth</i> : 20, <i>min_samples_leaf</i> : 5, <i>max_features</i> : square root, and <i>splitter</i> : best
RF	7	Obese rate, Overweight rate, Active transportation rate, Rate of active population, Log median age, Visible minority rate, Daily smokers rate	<i>max_depth</i> : 10, <i>min_samples_leaf</i> : 3, <i>max_features</i> : log2, and <i>n_estimators</i> : 150
XGBoost	6	Obese rate, Overweight rate, Rate of active population, Visible minority rate, Daily smokers rate, Recent immigrant rate	<i>max_depth</i> : 5, <i>min_samples_leaf</i> : 10, <i>n_estimators</i> : 150, and <i>learning_rate</i> : 0.1

Table 4. Summary of final feature sets (determined by RFECV) and model parameters (determined by grid search with 5-fold cross-validation) across the seven machine learning models. ^[1] *Regularization Parameter*, ^[2] *Same optimal feature set as LR model*, ^[3] *Multi-Layer Perceptron*, ^[4] *Radial Basis Function*

to each trained model, obtaining both local (neighbourhood-level) and global (model-level) explanations by quantifying each feature's contribution to the prediction⁴⁹. Table 4 summarizes the final features retained by each model and the optimal hyperparameters determined through grid search. Interestingly, certain features recurred across multiple top-performing models, including the modifiable behavioral and environmental variables (obesity and overweight rates, active transportation usage, overall physical activity level, and daily smoking rate) and the non modifiable demographic composition variables (visible minority rate and recent immigrant rate), all of which act as strong contextual predictors of diabetes prevalence across models. The modifiable variables are actionable targets for population level interventions, whereas the demographic variables can help identify priority populations and guide the tailoring of interventions to their specific needs.

All models were implemented in Python 3.11 using Scikit-learn 1.6.1. The code and trained model artifacts are available via our GitHub repository [<https://github.com/HIVE-UofT/diabetes-analysis>], and an interactive dashboard has been deployed for exploration of the results [<https://nediaml.hivelab-uoft.ca/>]. We exported trained models using PyMilo [<https://github.com/openscilab/pymilo>], which saves each model's parameters in a secure, non-executable JSON format to ensure cross-platform compatibility and to mitigate security concerns associated with binary serialization, an important consideration in healthcare applications.

Causal ML estimation through Causal Forest

Causal-ML methods move beyond correlation by estimating how a change in a specific factor would alter the outcome^{99,100}. While randomized controlled trials are the gold standard for establishing causality, conducting an RCT at the neighbourhood-level (e.g., randomly altering neighbourhood conditions) is often infeasible. Instead, we analyze observational data with causal-ML techniques that adjust for confounding and approximate experimental conditions. After controlling for the covariates (features) listed in Table 3, we assume that the treatment assignment is effectively independent of the diabetes rate each neighbourhood would otherwise have experienced. Under this assumption, and further assumptions about the temporal ordering of the variables in the cross-sectional setting, the estimated contrasts can be interpreted as effects of hypothetical interventions.

We work with the same set of neighbourhoods, $\mathcal{D} = \{(x_i, y_i)\}$, and introduce a binary treatment indicator $t_i \in \{0, 1\}$ for each area; a value of $t_i = 1$ denotes the presence of a favourable condition or intervention, whereas $t_i = 0$ denotes its absence. The potential outcomes $Y(0)$ and $Y(1)$ represent the diabetes prevalence under control and treatment, respectively. The Conditional Average Treatment Effect (CATE) for neighbourhoods characterized by $X = x$ is $\tau(x) = \mathbb{E}[Y(1) - Y(0) | X = x]$, the expected change in prevalence obtained by switching from control to treatment for areas with profile x .

We implemented Causal Forest using Microsoft Research's EconML library, which provides an adaptation of Generalized Random Forests for heterogeneous treatment effect estimation¹⁰¹. Causal Forest partitions the feature space such that neighbourhoods with similar characteristics but different treatment statuses can be compared in a quasi-experimental fashion. The algorithm thus finds subgroups of neighbourhoods where an intervention, such as increasing access to green space or healthy-food outlets, has a consistently different effect. By estimating both $Y(0)$ and $Y(1)$ for every neighbourhood, we obtain modeled counterfactual differences that can prioritize hypotheses about where public-health measures might be most beneficial, subject to the assumptions above.

We used a total of 10,000 trees in each Causal Forest to stabilize the treatment effect estimates. Model parameters (e.g., tree depth and minimum samples per leaf) were tuned via 10-fold cross-validation on the training data. For the outcome and treatment models inside the Causal Forest, we used a Lasso regression as the base learner. The Lasso imposes an L_1 regularization penalty that encourages sparsity. This regularization reduces over-fitting by selecting a concise set of predictors and yields more reliable estimates of $Y(0)$, $Y(1)$, and the treatment propensity, an important consideration for a high-dimensional, observational dataset.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to portions of the input data being licensed (DemoStats) or subject to strict third-party access controls (CCHS), but are available from the corresponding author on reasonable request. This study draws on three census-tract-level sources. ON-Marg (2021) is publicly available; DemoStats (Environics Analytics, 2022) is licensed and cannot be redistributed by the authors; Canadian Community Health Survey (CCHS) indicators are available as public-use microdata and, where finer detail is required, via Statistics Canada Research Data Centres under their access policies. All analysis code, variable definitions, and processing scripts are available at <https://github.com/HIVE-UofT/diabetes-analysis>, along with de-identified derived outputs. Researchers with appropriate access to DemoStats and CCHS can reproduce the analysis using our scripts.

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Author contributions

M.N., A.R., and I.G. jointly designed and implemented the machine learning pipeline, conducted the data analysis, and drafted the manuscript. M.N. and A.R. led the interpretation and analysis of results by developing the causal ML framework and creating visualizations. A.R. contributed to the web platform development. Z.S. provided supervision and contributed to data visualization. O.S., K.K., J.R.B., and Z.S. provided critical feedback throughout the revision process. All authors reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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